

Detailed Notes: Lectures 16–20

Introduction to Internet of Things

Based on the lecture PDFs on sensor networks, UAV networks, and machine-to-machine communication.

Lecture 16 — Sensor Networks III: Applications

This lecture moves away from basic sensor-network definitions and focuses on real applications. The main themes are target/object tracking, agricultural monitoring and intrusion detection, soil sensing for irrigation, and multimedia sensor networks.

1) Target tracking / object tracking

- Sensor networks are useful for surveillance in places such as railway stations, plazas, markets, and other crowded public spaces.
- The goal is either to track the target in real time as it moves or to predict its future trajectory.
- Two tracking styles are highlighted:
- Real-time tracking: follow the object continuously as it moves from one sensed region to another.
- Trajectory prediction: estimate the next location or a sequence of future positions from the current movement pattern.
- The lecture notes that target tracking is a major research problem because many applications involve multiple objects, not just one.
- Multi-object tracking is harder than single-object tracking, and efficient methods are needed instead of tracking every object independently.

2) Formulations of target tracking

- Push-based formulation: sensor nodes compute the target position and periodically notify the sink node.
- Poll-based formulation: nodes are periodically queried to report whether a target is present in their sensing range.
- Guided formulation: a tracker follows the trail of the target and intercepts it.
- These formulations differ in how information is generated, forwarded, and collected.

3) Cooperation, localization, prediction, and energy management

- Tracking requires cooperation among sensor nodes for data aggregation and relaying.
- Target position can be estimated using trilateration when the positions of multiple sensor nodes are known.
- After localization, prediction estimates where the target will be after a time interval Δt .
- Energy management is critical: only the minimum required sensor nodes should stay active, while the rest can go to sleep mode.

- The lecture stresses that tracking and energy saving must be designed together.

4) Agricultural applications

- Wireless sensor networks can monitor agricultural fields for intrusions and soil conditions.
- The field can be protected from intrusions by cattle or other animals that damage crops.
- The lecture presents AID, a prototype for Agricultural Intrusion Detection using Wireless Sensor Networks.
- AID uses two sensor types: passive infrared (PIR) sensors and ultrasonic sensors.
- PIR detects the presence of an intruder, while the ultrasonic sensor estimates the distance of the object from the sensor.
- When an intrusion is detected, the farmer can be notified through SMS or other alerts.

5) Layered architecture of agricultural intrusion detection

- Sensing layer: PIR and ultrasonic sensing.
- Processing layer: microcontroller-based processing.
- Routing layer: forwards the information through the network.
- Application layer: sends SMS and alert messages to the farmer or stakeholder.

6) Soil monitoring and irrigation

- The lecture also discusses sensing soil parameters such as moisture content, water level, and mineral content such as phosphorus and nitrogen.
- These measurements can be used to decide when irrigation is required.
- If soil moisture drops below a desired level, the field can be irrigated automatically or semi-automatically.

7) Multimedia sensor networks

- Multimedia sensor networks use multimedia devices in addition to regular scalar sensors.
- They can combine cameras with traditional sensors to support richer sensing and surveillance.
- The lecture indicates that camera activation can be coordinated depending on where an object is moving.
- Applications include surveillance, wildlife/habitat monitoring, and environmental monitoring.

Lecture 17 — Sensor Networks IV: Coverage

This lecture explains the coverage problem in wireless sensor networks and why coverage is central to sensor deployment. It also connects coverage to connectivity and presents the OGDC area-coverage algorithm.

1) What coverage means

- Coverage means deploying sensor nodes so that no point in the region of interest is left unsensed.

- In area coverage, each point of the area must lie within the sensing range of at least one sensor node.
- Coverage is closely related to connectivity: sensed data must also be able to reach the sink node.

2) Sensing range vs transmission range

- Sensing range is the radius within which a node can detect events.
- Transmission range (communication range) is the radius within which a node can communicate.
- Typically, the transmission range is larger than the sensing range.
- A key result attributed to Zhang and Hou states that if $R_c \geq 2R_s$, then coverage implies connectivity.
- This means that once coverage is ensured, connectivity is automatically satisfied under that condition.

3) Coverage goals and deployment strategies

- The objective is to monitor or track the sensing field efficiently using a minimum number of sensors.
- For static sensors, the key problem is where to deploy or activate them.
- For mobile sensors, the key problem is how to plan the sensor trajectory.
- Coverage algorithms may be centralized, distributed, or localized.
- Sensors can also be deployed deterministically or randomly.

4) Reporting models and network lifetime

- Event-driven reporting: data is sent when an event occurs, such as forest fire or building fire monitoring.
- On-demand reporting: data is sent in response to a query, such as in inventory systems.
- A core objective is to maximize network lifetime while using the smallest possible sensor set.
- Network lifetime can be defined in different ways: until the first node dies, until the last node dies, or until a given fraction of nodes fail.

5) Major types of coverage

- Area coverage: the most common form; every point in an area must be sensed.
- Point coverage: a specific set of points must be covered, often with minimum sensors.
- Barrier coverage: a boundary or strip must be monitored so that intruders cannot pass unnoticed.

6) Barrier coverage in detail

- One-barrier coverage means at least one sensor detects an intrusion.
- Two-barrier coverage means at least two sensors detect it.
- k-barrier coverage means at least k sensors detect it.
- The lecture contrasts weak coverage and strong coverage.
- Weak coverage leaves possible paths for intruders to pass undetected, while strong coverage blocks all such paths.

7) Crossings and the OGDC algorithm

- A continuous region is covered if crossings exist and every crossing is covered.
- Crossings are formed by intersections between circles or between a circle and the region boundary.
- The OGDC algorithm aims to cover crossings with minimum overlap and minimum sensor use.
- Nodes become active in positions that approximate an optimal geometric layout.
- A common optimal spacing uses equilateral-triangle geometry, with distance $\sqrt{3} \times R_s$ between selected active nodes.
- A node that is fully covered can go to sleep mode.
- The process repeats round by round until the full area is covered.

Lecture 18 — Sensor Networks V: Mobile Sensor Networks

This lecture introduces mobility in sensor networks and explains why mobile elements can solve problems that static deployments cannot, especially partitioning and limited adaptability.

1) Static vs mobile sensor networks

- Static or stationary sensor nodes do not move after deployment.
- Static networks are easier to deploy and maintain, but failures can partition the network.
- A failed critical node can split the network into multiple fragments.
- Topology cannot change automatically in a stationary network.

2) Why mobility helps

- Mobile nodes can move to replace failing nodes or to reconnect fragmented parts of the network.
- Mobility allows the network to reorganize itself dynamically.
- Mobile wireless sensor networks conceptually combine wireless sensor networks with MANET-like behavior.

3) Key elements of mobile wireless sensor networks

- Mobile sensor nodes can move and sense physical parameters from the environment.
- Mobile sink: the sink itself moves to collect data from stationary sensor nodes.
- Data mule: an external mobile entity collects data from sensors and delivers it to the sink.
- These models help when direct communication to the sink is not always possible.

4) Environments and forms of mobility

- Terrestrial mobile sensor networks: nodes on land, often linked with robots or aerial support.
- Aerial mobile sensor networks: nodes fly and sense, then send data to a surface station.
- Underwater mobile sensor networks: nodes mounted on underwater vehicles or similar platforms for marine monitoring.

5) Examples of mobile entities

- Humans: mobility is unpredictable, which creates challenges for sensing and data delivery.
- Vehicles: sensors in cars and modern vehicles form vehicular sensor networks.
- Mobile robots: controllable nodes that collect data based on predefined instructions.

6) Human-centric sensing

- Smartphones and PDAs enable participatory sensing, people-centric sensing, and opportunistic sensing.
- Participatory sensing: many users actively turn on sensor-equipped phones to collect data.
- People-centric sensing: humans themselves are treated as sensing participants.
- Opportunistic sensing: sensing and reporting happen whenever a node happens to move into a useful location or proximity.

Lecture 19 — UAV Networks

This lecture discusses networks formed by flying objects such as drones and quadcopters. The focus is on topology, control, constraints, advantages, and the relationship to flying ad hoc networks.

1) What UAV networks are

- A UAV network is a network of flying objects, especially drones or quadcopters, that communicate with one another.
- Applications include civilian monitoring, post-disaster assessment, border surveillance, and military reconnaissance.

2) Core features

- Topology changes very quickly because UAVs move at high speed.
- Popular topologies include star and mesh.
- SDN is discussed as a way to flexibly manage services and deployment.
- Routing must be adaptive and energy-aware.
- UAVs can support multitasking and large-area coverage.
- Networks are reconfigurable for different missions.

3) Key issues and constraints

- Relative positions of UAVs change rapidly.
- Malfunctioning UAVs must not break the whole network.
- Links are intermittent because the geometry of the network keeps changing.
- Traditional routing approaches from ground networks may not work well.
- UAVs face high power demands because they must fly, carry payloads, and communicate.
- The environment adds stress through wind, rain, and similar effects.

4) Single UAV vs multi-UAV systems

- Single UAV systems have limited scalability and lower survivability.
- Multi-UAV systems are more scalable, faster for missions, and more survivable.
- The trade-off is higher coordination complexity and often higher cost.

5) Topology comparison

- Star topology: UAVs connect to a ground station; point-to-point communication is emphasized.
- Multi-star topology: several star clusters connect to the ground station.
- Mesh topology: multi-point to multi-point communication, with greater fault tolerance and scalability.

6) Flying Ad Hoc Networks (FANETs)

- When UAV-based ad hoc networking is viewed as a flying version of MANETs, it is called a FANET.
- Communication can be inter-plane, intra-plane, ground-station, or ground-sensor communication.
- Air-to-air links can use wireless interfaces such as ZigBee / IEEE 802.15.4 or Wi-Fi / IEEE 802.11.
- UAVs can also support collaborative sensing and data muling.

7) Applications

- Disaster monitoring and post-disaster response.
- Emergency network establishment.
- Surveillance and reconnaissance.
- Environment and border monitoring.

Lecture 20 — Machine-to-Machine (M2M) Communication

This lecture explains machine-to-machine communication as a core building block of IoT. The emphasis is on autonomous behavior, cross-platform communication, ecosystem roles, and node/device classes.

1) What M2M means

- M2M is communication between machines or devices with communication and computing capability.
- The goal is to minimize or eliminate human intervention.
- It supports autonomous operation in IoT systems.

2) Simple examples

- A smart home robot can open a refrigerator, take out milk, and pour it into a glass without human help.
- If milk is missing, the robot or refrigerator can automatically send an SMS to the milk supplier.

- In a highway accident scenario, sensors on cars can trigger an alert that reaches remote servers, hospitals, ambulances, and doctors.
- Humans may receive the information, but they do not manually operate the network.

3) Relationship with SCADA

- M2M is compared conceptually with SCADA.
- SCADA is traditionally wired and often designed for isolated proprietary systems.
- M2M is presented as a wireless, cross-platform approach that connects heterogeneous devices and technologies.

4) Typical M2M pipeline

- Sensors produce data.
- Information is extracted and processed.
- If needed, actuators perform an action, such as opening a valve in agriculture when soil moisture drops.

5) M2M application areas

- Environmental monitoring.
- Civil protection and public safety.
- Supply chain management.
- Energy and utility distribution / smart grid.
- Intelligent transportation systems.
- Healthcare.
- Building automation.
- Military applications.
- Agriculture and home networks.

6) Features of M2M

- Large number of devices.
- Low cost and energy-efficient nodes.
- Small traffic per machine, but large collective data volume.
- Strictly minimal human intervention; humans are needed mainly for operational stability and sustainability.

7) Node classes

- Low-end nodes: cheap, low-capability, heavily resource constrained, high-density deployment, often used for environmental monitoring.
- Mid-end nodes: more expensive, some mobility, better functionality, support for localization, QoS, TCP/IP, and control intelligence.
- High-end nodes: low-density deployment, can handle multimedia/video, often mobile, used in military or biomedical applications.

8) M2M ecosystem and platform architecture

- Main actors include device providers, internet service providers, platform providers, service providers, and service users.

- Data often flows from an M2M area network through a gateway to the internet, then to a platform provider and onward to services/users.
- RESTful interfaces are used because they work well in resource-limited environments.
- The M2M service platform supports device profile management, device/network management, device searching, user management, authentication, charging, service management, connection management, app management, and web portal access.
- The device platform maintains device profiles such as location, type, address, and description, and it monitors status and controls devices accordingly.

End of notes.